

Task 1 - List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

SMPS vs UPS devices

SMPS devices (convert AC mains to stable DC efficiently, with minimal heat/size):

1. **Desktop PC** – needs multiple regulated DC voltages (+3.3V, +5V, +12V) efficiently for CPU, motherboard, and drives.
2. **LED TV/Monitor** – needs compact, low-heat power conversion since it's housed in a thin chassis.
3. **Laptop charger (power brick)** – needs to be lightweight and portable while converting AC to the DC voltage the battery requires.

UPS devices (need uninterrupted power during outages):

1. **College server/networking rack** – a sudden power cut would crash servers and corrupt data, so a UPS bridges the gap until backup generators kick in.
2. **Desktop PC** – an abrupt shutdown during a power cut can cause unsaved work loss or file corruption, so a UPS gives time to save and shut down safely.
3. **CCTV Security Systems and Digital Video Recorders (DVRs)** - Security systems must remain operational 24/7. Intruders could deliberately cut the power, or a standard power loss could leave a facility completely unmonitored and vulnerable without backup power.